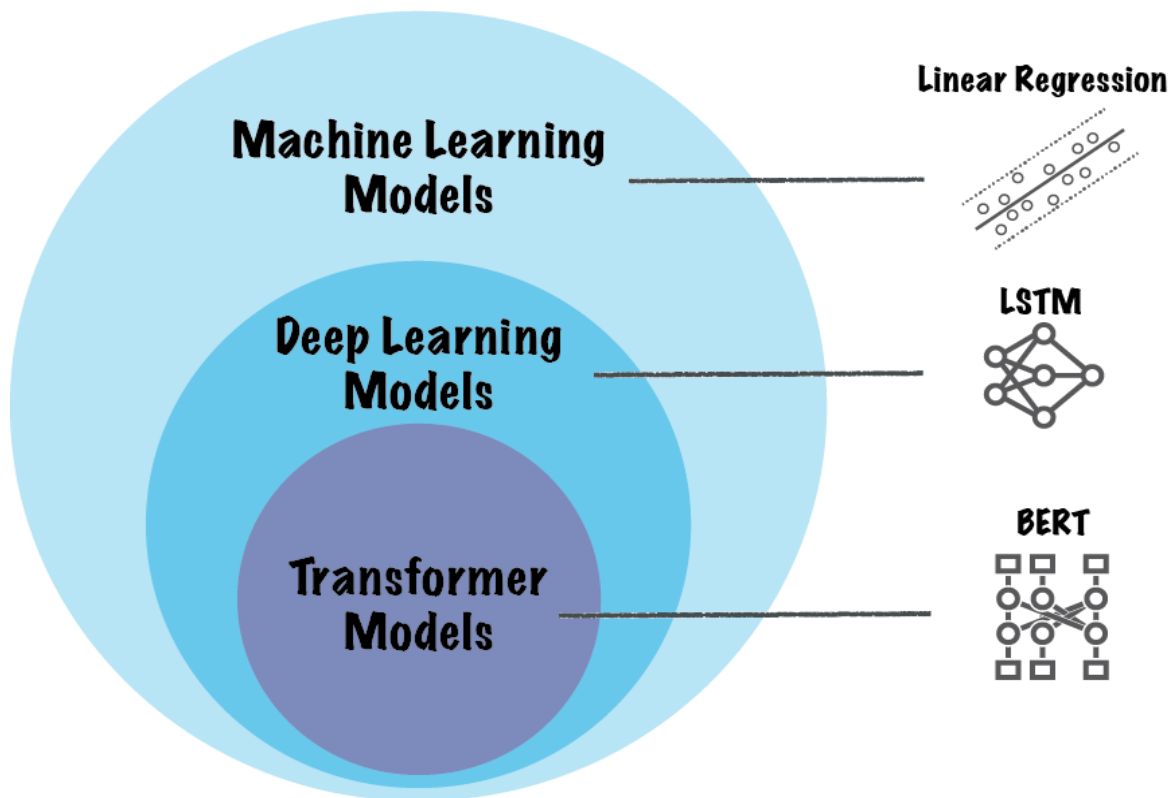


TRANSFORMERS' EXPLANATION TO A 7TH STANDARD KID



What Are Transformers (In AI)?

Transformers are a type of computer model that help computers understand and work with human language — like English, Hindi, or any other language!

They are used in AI tools like ChatGPT, Google Translate, and Siri.

Imagine This...

You're reading a sentence:

"The cat sat on the ____."

You already *know* the missing word is "mat", right?

That's because your brain remembers the words before it and understands the meaning.

Well, computers used to be really bad at that. But Transformers changed everything!

A Short History

-  **2017: A big team at Google created a new AI model called a Transformer.**
 -  **They published a paper titled: "Attention is All You Need."**
 - **This was a *big deal* because this model could:**
 - **Read long texts**
 - **Translate languages better**
 - **Answer questions smartly**
 - **And even write stories!**
-

How Does a Transformer Work?

Let's explain it with 4 key ideas.

1. Tokens Are Like Puzzle Pieces

- **First, a sentence is broken into tokens (small pieces of words).**

- **Example: “I love apples” → [“I”, “love”, “apples”]**
-

2. Embeddings = Turning Words into Numbers

- **Computers don’t understand words. They understand numbers.**
- **So each token becomes a special set of numbers called an embedding.**

It's like giving every word its own secret code!

3. Attention – The Magic Trick! 🎩

Here comes the superhero of Transformers: Attention

💡 **It helps the AI focus on the right words when reading a sentence.**

Example:

“The dog chased the cat because it was scared.”

Who was scared? The dog or the cat?

With attention, the AI can figure out that “it” refers to “the cat”, because of the sentence meaning.

🧠 **The AI is "paying attention" to different words when trying to understand the sentence.**

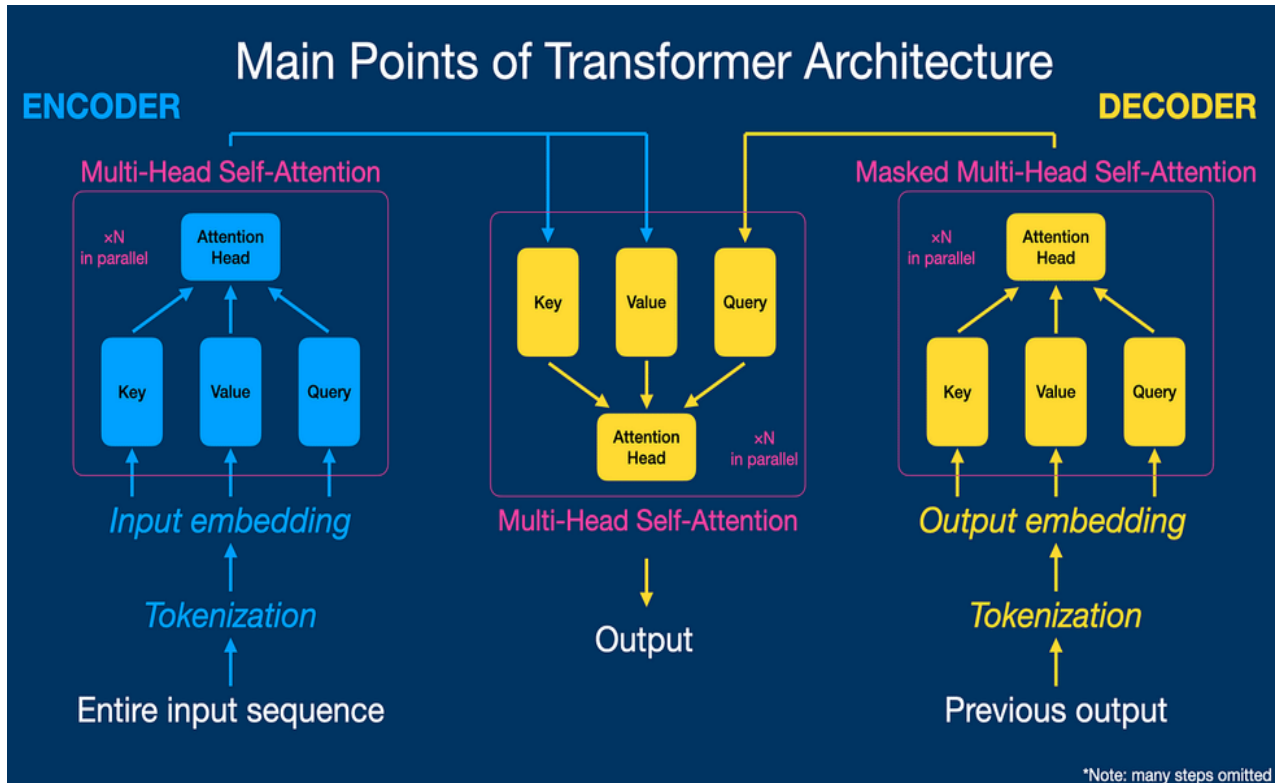
4. Layers – Like a Big Cake 🍰

Transformers have many layers, like a tall cake.

- **Each layer learns something new about the sentence.**

- **The more layers, the smarter the model becomes.**

Big models like ChatGPT have dozens or even hundreds of layers!



🔧 What Can Transformers Do?

Here are some cool things they can do:

🧠 Task

Translation

Text summarization

Chatting with users

Answering questions

Writing stories or poems

💡 Example

English → French

Make a long article short

Like what we're doing now!

“Who is the president of India?”

“Write a poem about space.”



Task

Coding help



Example

“Fix my Python code.”



Fun Fact:

Transformers are the reason AI got super smart in the last few years.

They’re used in:

- **ChatGPT (by OpenAI)**
 - **BERT and T5 (by Google)**
 - **LLaMA (by Meta)**
 - **Claude (by Anthropic)**
-



Summary (Just Like in School!)

- **Transformers help computers understand language.**
- **They use attention to figure out which words are important.**
- **They turn words into numbers (embeddings).**
- **They go through layers to get smarter and better.**
- **They power tools like ChatGPT that can talk, translate, and help with homework!**



Comic Title: "The Secret Life of Transformers (Not the Robots!)"



Panel 1: "Breaking It Down"



A cartoon student says:

"I want to teach my computer to understand this sentence:
'The bird flew over the house.'"



The sentence gets *snipped* into little word bubbles:

- "The"
- "bird"
- "flew"
- "over"
- "the"
- "house"



Label:

Step 1: Split the sentence into tokens!



Panel 2: "Words Become Numbers!"

A friendly robot scientist stands in front of a computer screen. Each word turns into a glowing code like:

- "bird" → [0.8, 0.2, 0.9]
- "flew" → [0.5, 0.7, 0.1]

💡 **The robot says:**

"Words are now embeddings — special number codes computers can understand!"

Label:

Step 2: Turn words into numbers!

🦱 **Panel 3: "Enter... Attention Man!"**

A superhero with a big “A” on his chest zooms in with a spotlight!

He shines his light on “bird” and “flew” and says:

"These words are most important to understand what’s going on!"

🌟 **Background shows other words dimmed out.**

Label:

Step 3: Attention helps the AI focus on important words!

🍰 **Panel 4: "Layer Cake Learning!"**

A huge, 5-layer chocolate cake appears!

Each layer has a little robot climbing up with a *thinking bubble*:

- **Layer 1: "Hmm... sentence structure!"**
- **Layer 3: "Aha! It's about flying!"**
- **Layer 5: "I understand the whole meaning!"**

 **Label:**

Step 4: Transformers use many layers to learn more at each step!

 **Panel 5: "Meet ChatGPT!"**

A cute, helpful robot with a smile (that's ME!) waves and says:

"Now I can answer questions, write stories, and help with homework!"

A student says:

"Wow! You're smart!"

Label:

Step 5: You're now chatting with a Transformer in action!

 **Final Box (Like a Lesson Recap)**



-  **Words → Tokens**
-  **Tokens → Numbers**
-  **Attention → Focus**
-  **Layers → Learning**
-  **Robot → Smart answers!**

 **The student gives a thumbs up:**

"Transformers aren't just robots. They're how AI understands me!"

